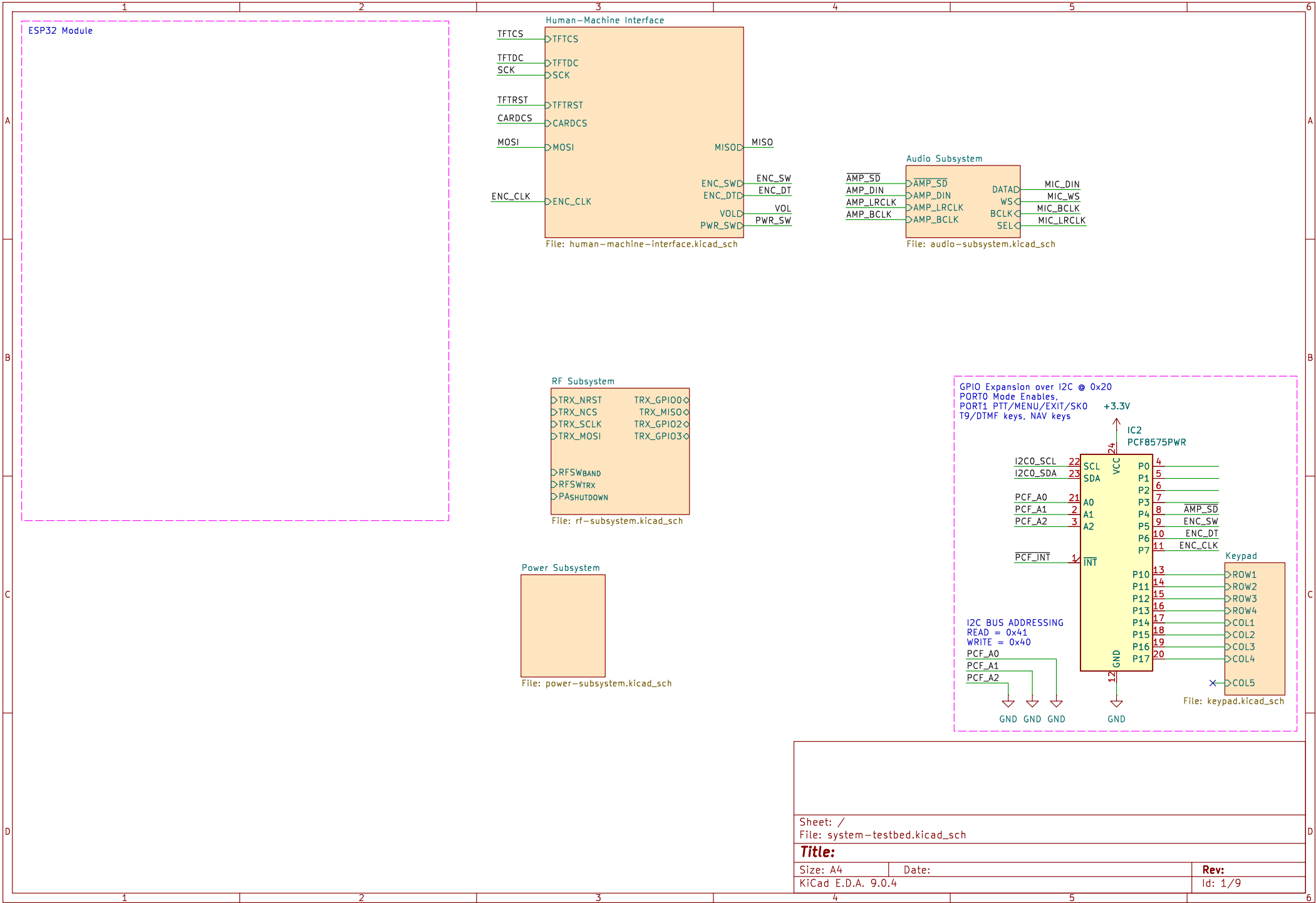
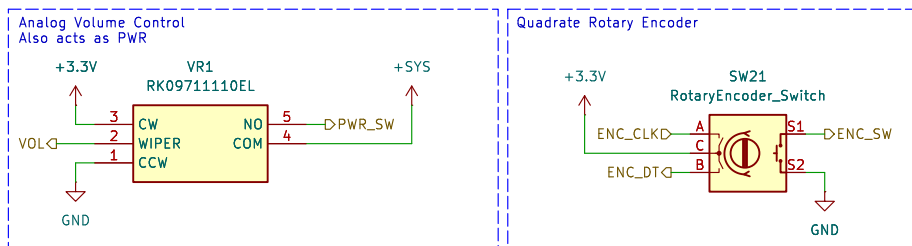
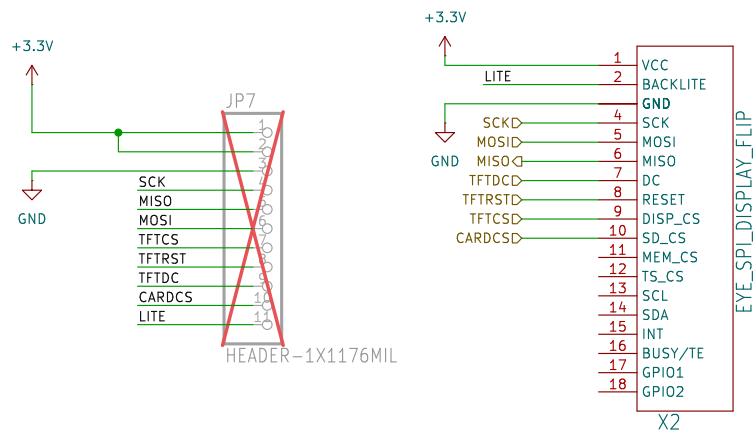






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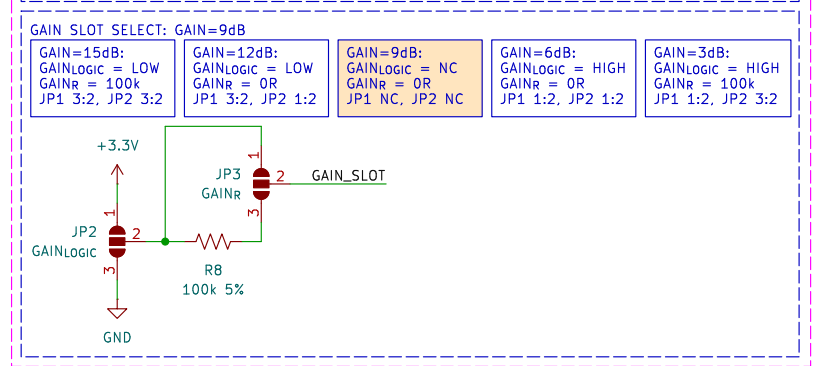
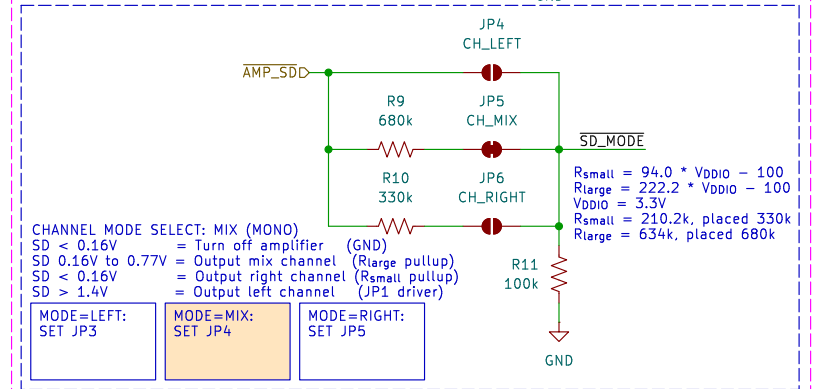
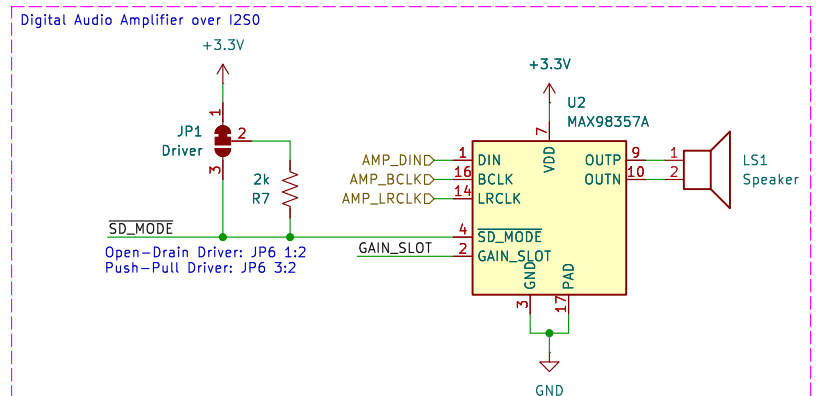
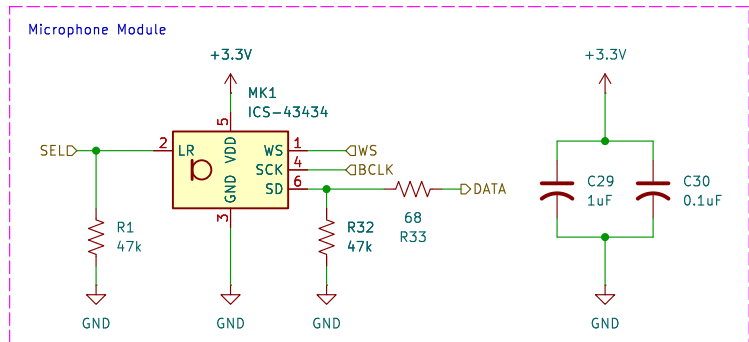
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Date:

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KiCad E.D.A. 9.0.4

Id: 2/9



Sheet: /Audio Subsystem/
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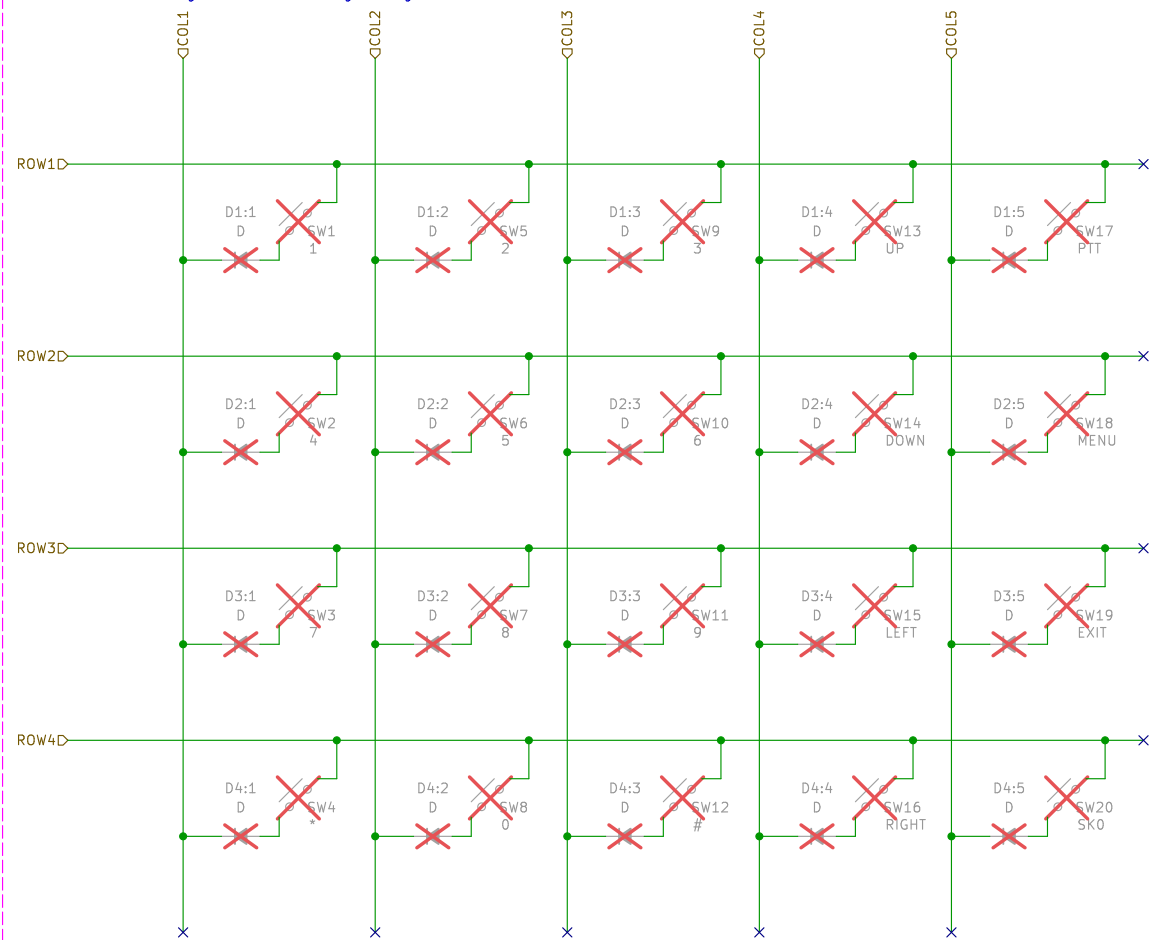
Size: A4 Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 3/9

Keypad Matrix
Application Note: The ROW pins get pulled up by the GPIO expander, with one ROW being pulled down.
The scanner relies on the COL pins being read in as a logic swing to low.
Remarked as Domino Logic, a rail-to-rail logic swing occurs.



Keypad Daughterboard

HW-834 Module

ROW4 1
ROW3 2
ROW2 3
ROW1 4 J1
COL1 5
COL2 6
COL3 7
COL4 8

H1 MountingHole_2.2mm_M2_Pad_Via
H2 MountingHole_2.2mm_M2_Pad_Via

Sheet: /Keypad/
File: keypad.kicad_sch

Title:

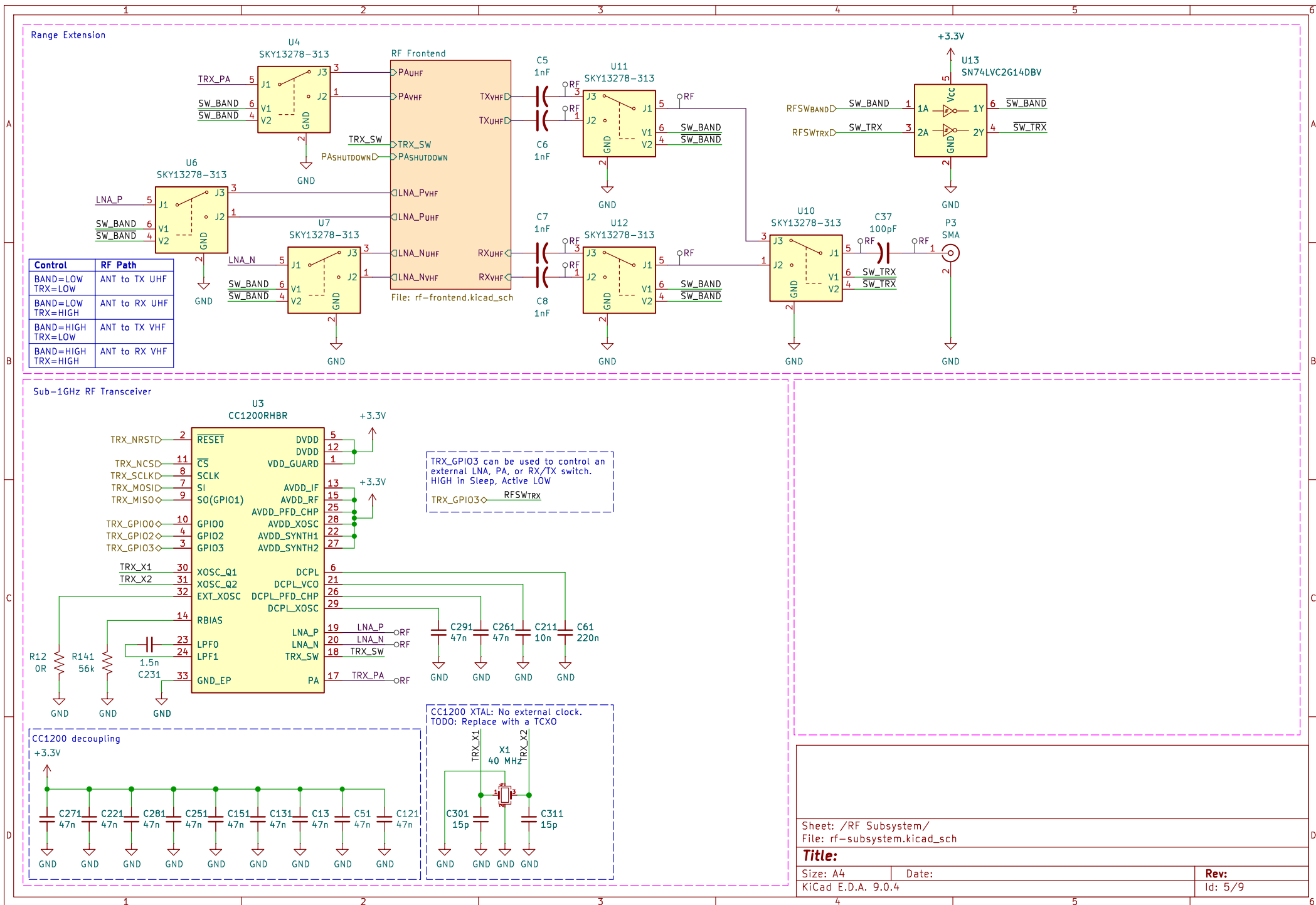
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KiCad E.D.A. 9.0.4

Rev:

Id: 4/9

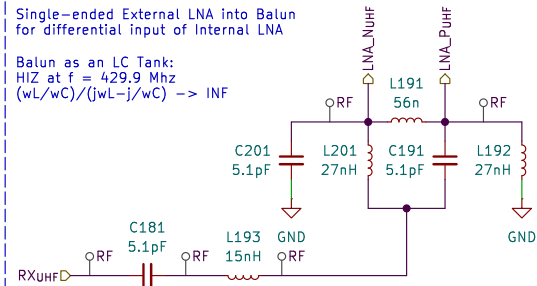


RF Frontend for UHF Reference 420-470 MHz Tune Separated RX and TX Signal Chains

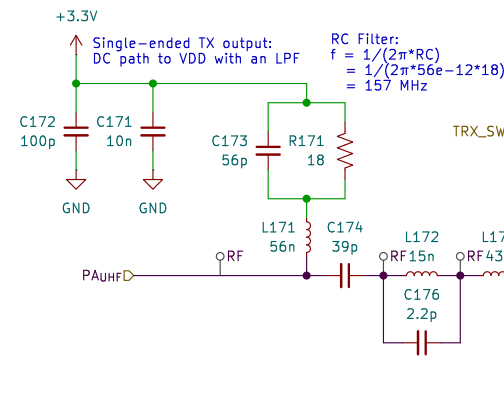
RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA

Balun as an LC Tank:
HIZ at $f = 429.9 \text{ MHz}$
(wL/wC)/($jwL-j/wC$) $\rightarrow \text{INF}$



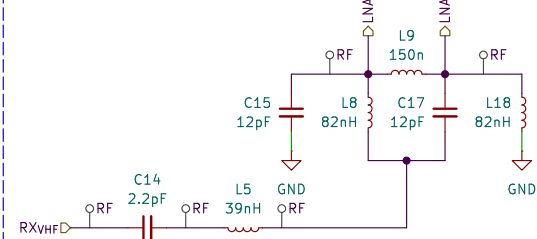
TX Signal Chain



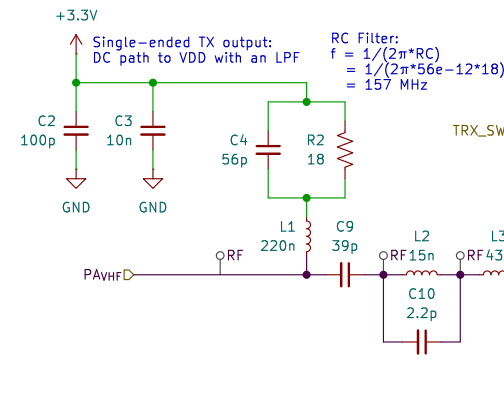
RF Frontend for VHF Reference 169-170 MHz Tune Separated RX and TX Signal Chains

RX Signal Chain

Single-ended External LNA into Balun for differential input of Internal LNA



TX Signal Chain



Sheet: /RF Subsystem/RF Frontend/
File: rf-frontend.kicad_sch

Title:

Size: A4

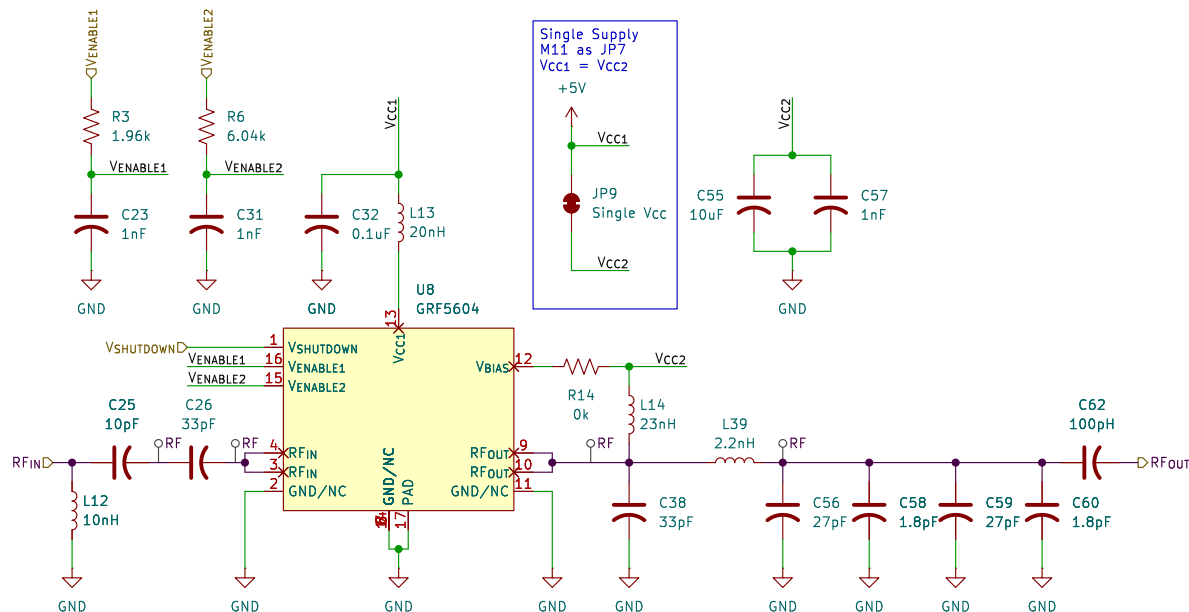
Date:

KiCad E.D.A. 9.0.4

Rev:

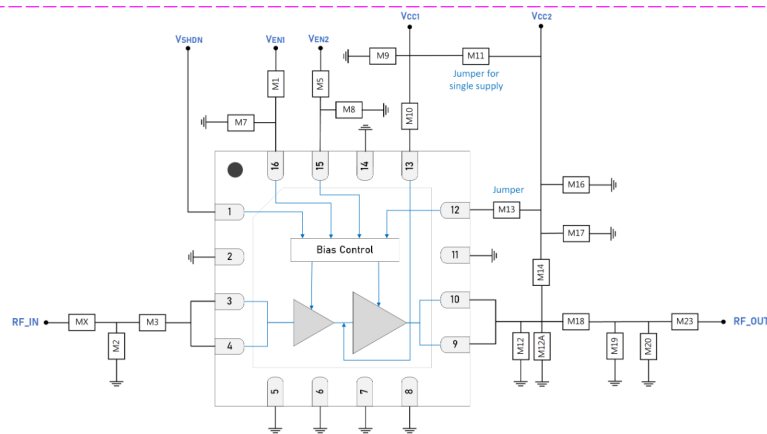
Id: 6/9

RF Amplifier for UHF Reference 390-470 MHz Tune



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 μ F must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/UHF 6W PA/
File: rf-power-amplifier-uhf.kicad_sch

Title:

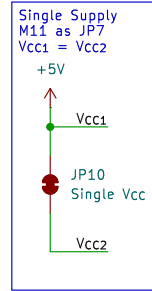
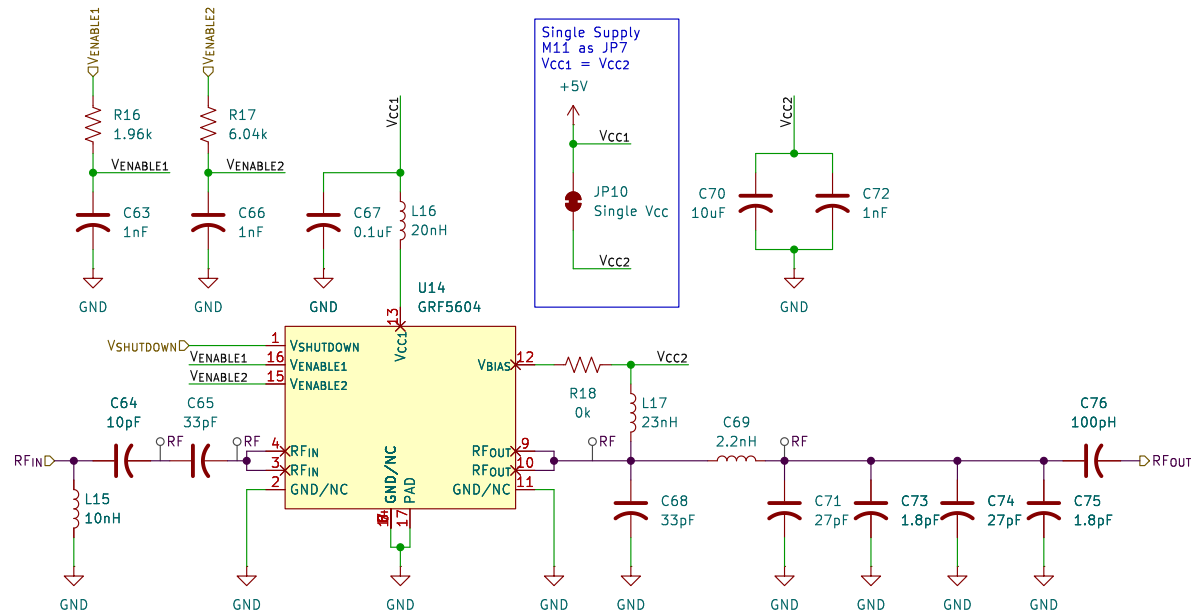
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KiCad E.D.A. 9.0.4

Date:

Rev:

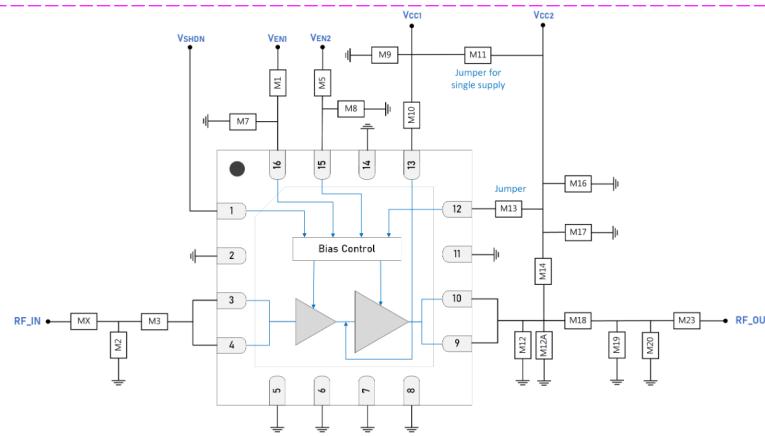
Id: 7/9

RF Amplifier for VHF
Needs unique VHF Tune?



Ref	Eval. Ref	450-470 MHz
R	M1	1.96k
L1	M2	10nH
C5	M3	33pF
C3	MX	10pF
R	M5	6.04k
C	M7	1000pF
C	M8	1000pF
C	M9	0.1uF
L	M10	20nH
R	M11	0k
C7	M12	33pF
C	M12A	2.0pF
R	M13	0k
L	M14	23nH
C	M16	**10uF
C	M17	1000pF
L	M18	2.2nH
C8	M19	27pF
C	M20	1.8pF
C	M23	100pF

**10 uF must be rated for > 5 V at maximum ambient temperature. Manufacturer Part Number in this case = GRM155C80J106ME11D.



GRF5604 Standard Evaluation Board Schematic

Sheet: /RF Subsystem/RF Frontend/VHF 6W PA/
File: rf-power-amplifier-vhf.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.4

Rev:

Id: 8/9

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G

H

Sheet: /Power Subsystem/
File: power-subsystem.kicad_sch

Title:

Size: CDate:KICad E.D.A. 9.0.4

Rev:Id: 9/9